Leibniz FH

Wirtschaftsinformatik (B. Sc.)
Modul 20.1 (Vertiefung Entwicklung)
Sommersemester 2026

Anwendung erweiterter Softwareentwicklung

2. Setup

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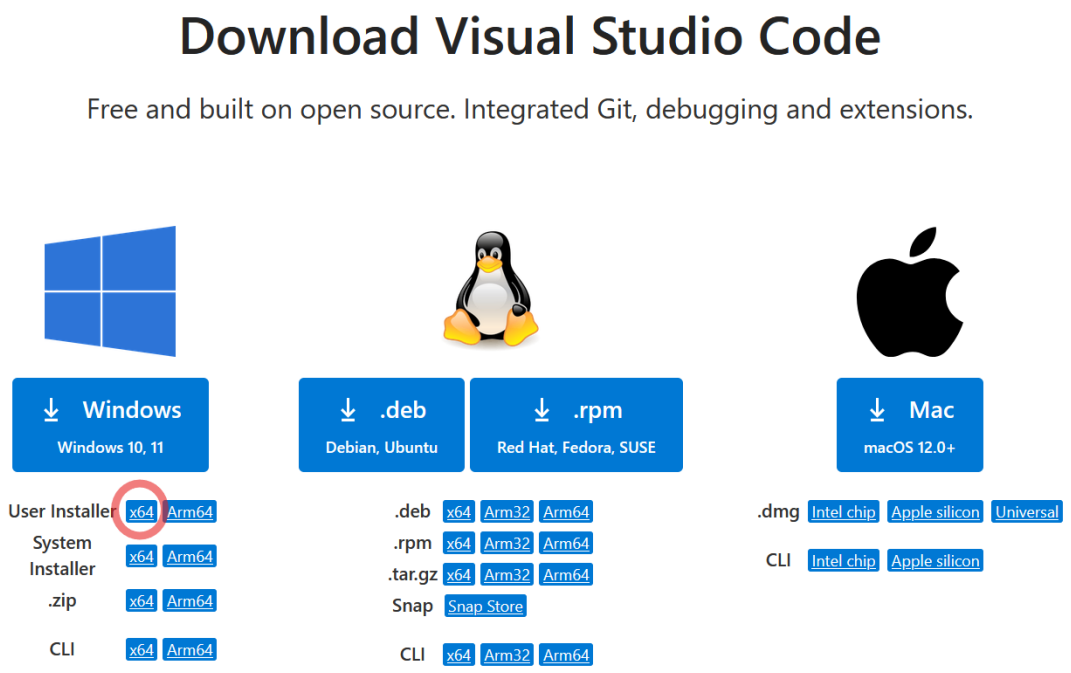
Die duale Hochschule für
Wirtschaft und Informatik in Niedersachsen

Downloads – VS Code

Direktlink Windows x64:
https://code.visualstudio.com/sha/download?build=stable&os=win32-x64-user

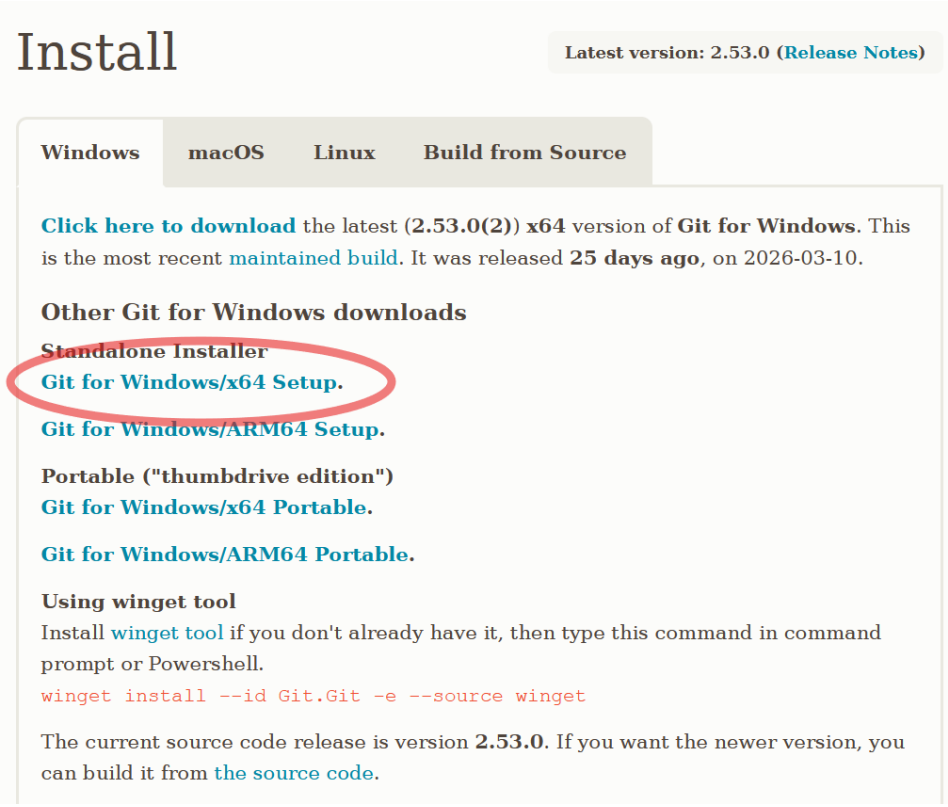
Download-Seite: https://code.visualstudio.com/download



Downloads - Git

Direktlink Windows x64:
https://github.com/git-for-windows/git/releases/download/v2.53.0.windows.2/Git-2.53.0.2-64-bit.exe

Download-Seite: https://git-scm.com/install/windows



Install

Latest version: 2.53.0 (Release Notes)

Windows | macOS | Linux | Build from Source

Click here to download the latest (**2.53.0(2)**) **x64** version of **Git for Windows**. This is the most recent maintained build. It was released **25 days ago**, on 2026-03-10.

Other Git for Windows downloads

Standalone Installer
Git for Windows/x64 Setup.

Git for Windows/ARM64 Setup.

Portable ("thumbdrive edition")
Git for Windows/x64 Portable.

Git for Windows/ARM64 Portable.

Using winget tool
Install winget tool if you don't already have it, then type this command in command prompt or Powershell.

```
winget install --id Git.Git -e --source winget
```

The current source code release is version **2.53.0**. If you want the newer version, you can build it from the source code.

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2.1. Komponenten

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IDE (Visual Studio Code)

- ✓ Editor, Terminal, Debugger in einem Tool
- ✓ weniger Brüche, mehr Überblick
- ✓ Autovervollständigung (KI Copilot) und Fehlermeldungen direkt im Code
- ✓ Standardwerkzeug für zahlreiche Programmiersprachen
- ✓ gute Alternative: PyCharm (JetBrains)

Terminal / Shell (Windows Terminal / Git Bash)

- ✓ Shell: Ausführungsumgebung und -kontrolle
- ✓ Terminal: Fenster/Container, in dem die Shell läuft
- ✓ Zugang zu Git, Python, Poetry und Co.
- ✓ präzise, automatisierbare Arbeitsabläufe
- ✓ wichtig für Server, Container, CI
- ✓ Bash ist (meist) Standard-Shell unter Linux
  - Zsh ist Erweiterung von Bash – Standard unter MacOS
  - Tolles Shell-Framework: Oh My Zsh

Versionskontrolle (Git)

- ✓ Änderungen nachvollziehbar speichern
- ✓ frühere Stände vergleichen und wiederherstellen
- ✓ risikoarm experimentieren
- ✓ Basis für Kollaboration und GitHub

Python Version Manager (pyenv)

- ✓ mehrere Python-Versionen parallel
- ✓ pro Projekt die passende Version
- ✓ weniger Konflikte mit globalen Installationen

Python Paket-Manager (Poetry)

- ✓ Abhängigkeiten sauber verwalten
- ✓ virtuelle Umgebungen pro Projekt
- ✓ reproduzierbare Setups auf anderen Rechnern

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2.2. Visual Studio Code

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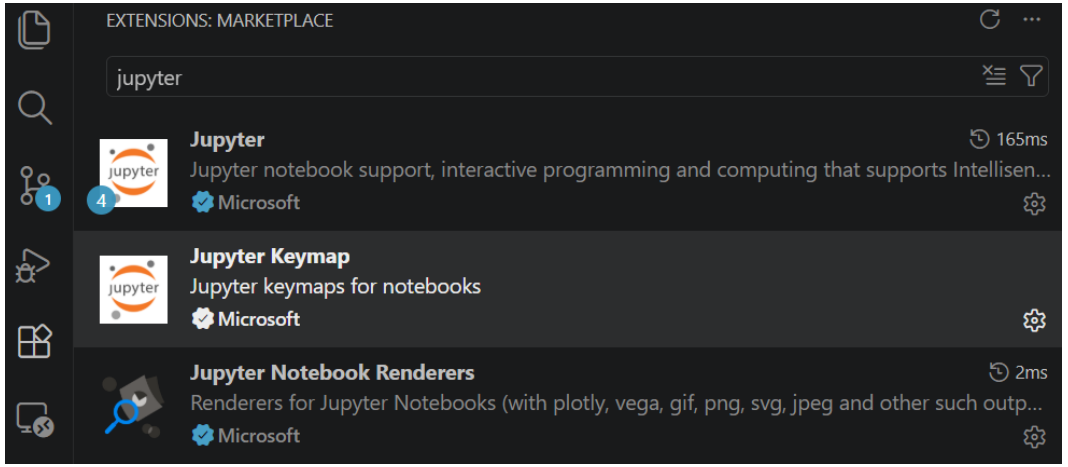
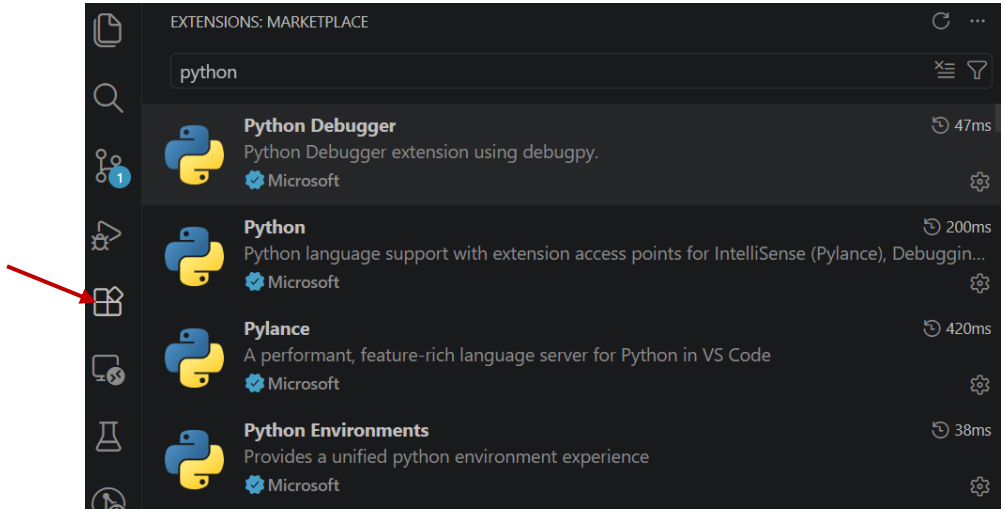
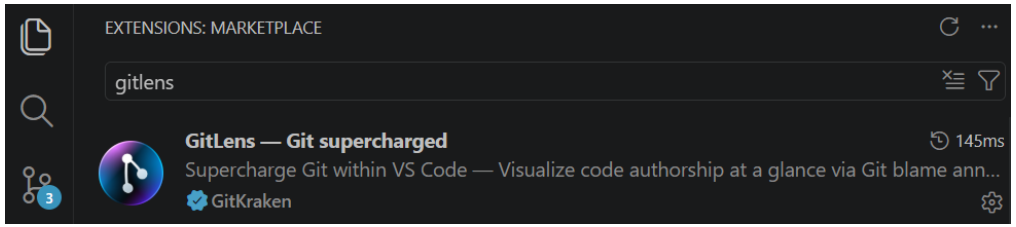
VS Code Installation

- ✓ Setup sollte ohne Admin-Rechte funktionieren
  - Bei Bedarf hier installieren:
    `C:\Users\<USERNAME>\AppData\Local\Programs`
- ✓ Extensions: Python, Jupyter, GitLens







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2.3. Git und Shell

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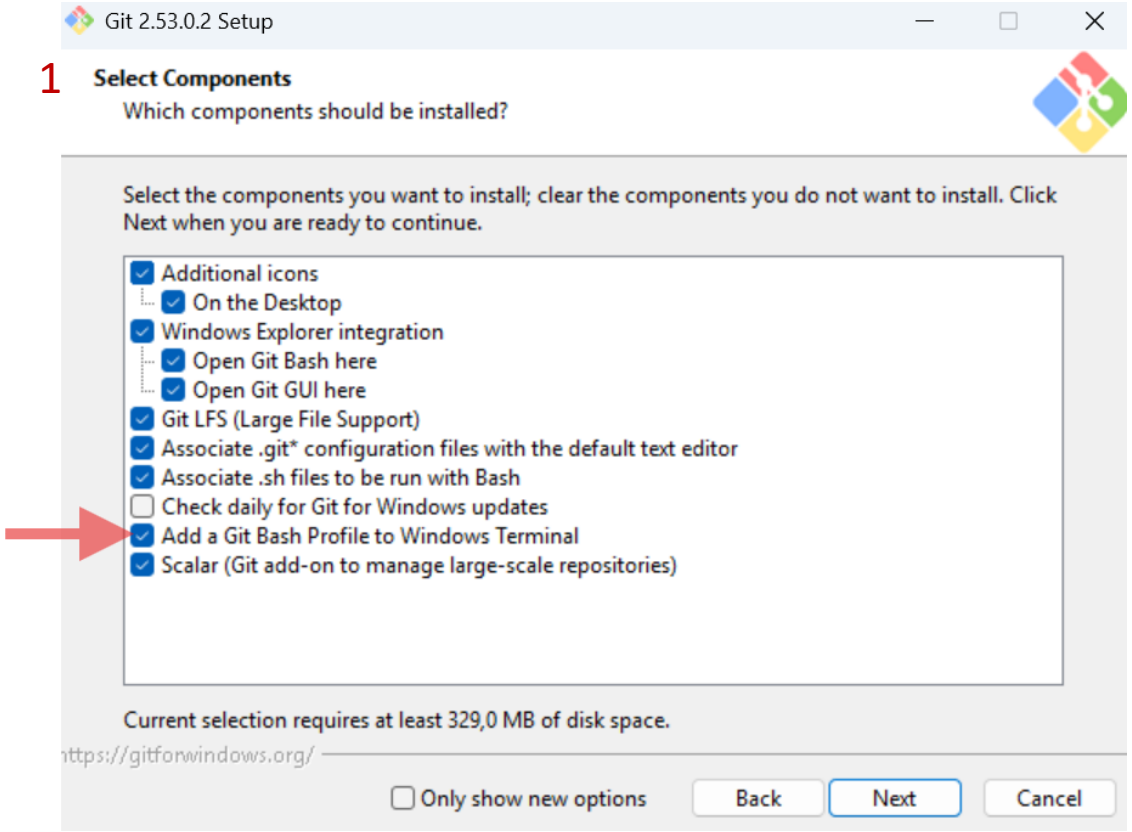
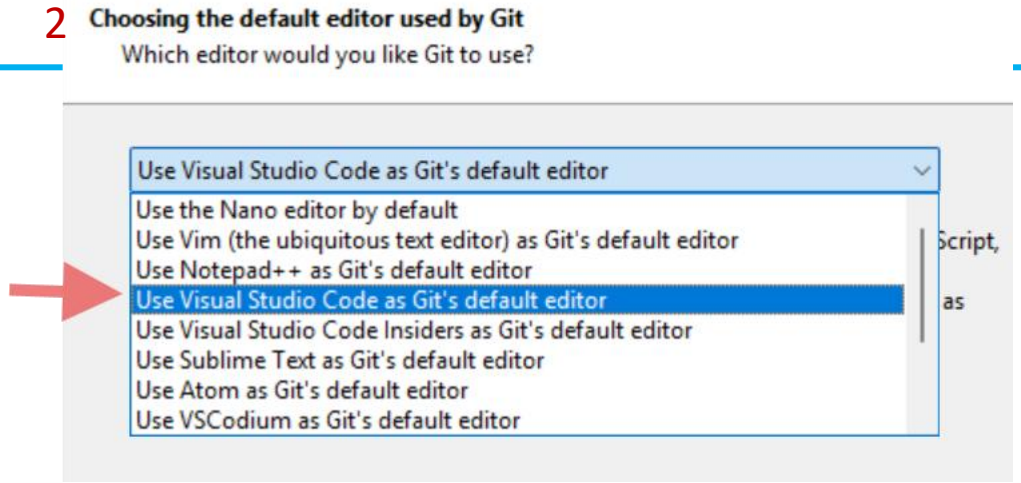
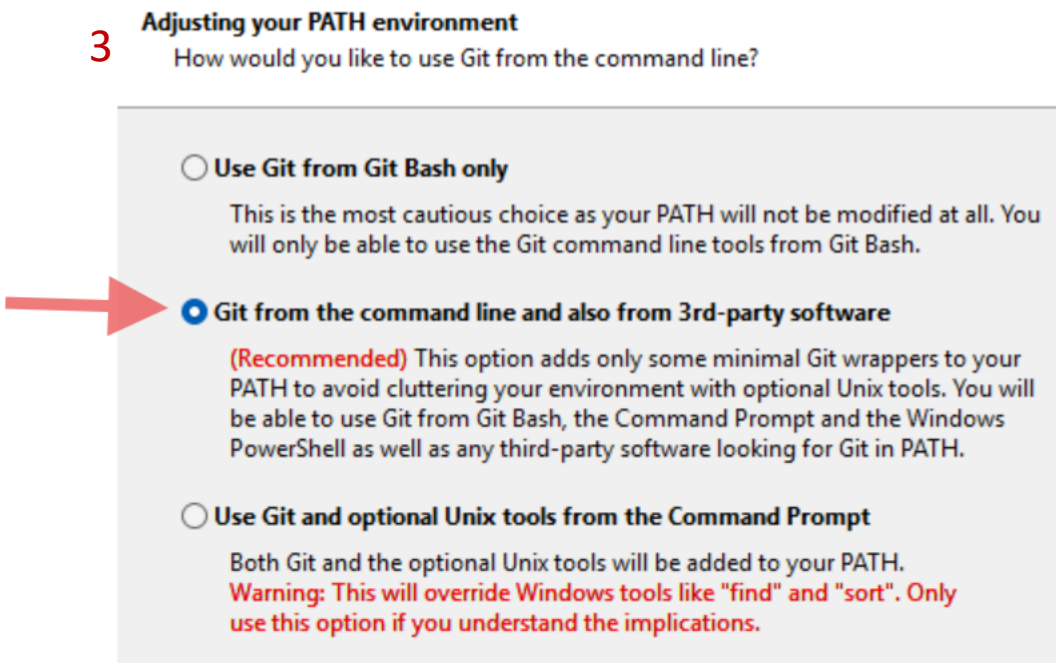
Git Installation

- ✓ Git for Windows installieren
  - Git Bash wird mitinstalliert







Git Installation .

4 **Choosing the SSH executable**

Which Secure Shell client program would you like Git to use?

- (●) **Use bundled OpenSSH**
  This uses ssh.exe that comes with Git.
- ( ) **Use external OpenSSH**
  This uses an external ssh.exe. Git will not install its own OpenSSH (and related) binaries but use them as found on the PATH.

5 **Choosing HTTPS transport backend**

Which SSL/TLS library would you like Git to use for HTTPS connections?

- ( ) **Use the OpenSSL library**
  Server certificates will be validated using the ca-bundle.crt file.
- (●) **Use the native Windows Secure Channel library**
  Server certificates will be validated using Windows Certificate Stores. This option also allows you to use your company's internal Root CA certificates distributed e.g. via Active Directory Domain Services.

6 **Configuring the line ending conversions**

How should Git treat line endings in text files?

- ( ) **Checkout Windows-style, commit Unix-style line endings**
  Git will convert LF to CRLF when checking out text files. When committing text files, CRLF will be converted to LF. For cross-platform projects, this is the recommended setting on Windows ("core.autocrlf" is set to "true").
- (●) **Checkout as-is, commit Unix-style line endings**
  Git will not perform any conversion when checking out text files. When committing text files, CRLF will be converted to LF. For cross-platform projects, this is the recommended setting on Unix ("core.autocrlf" is set to "input").
- ( ) **Checkout as-is, commit as-is**
  Git will not perform any conversions when checking out or committing text files. Choosing this option is not recommended for cross-platform projects ("core.autocrlf" is set to "false").

Git Installation .

7 **Configuring the terminal emulator to use with Git Bash**

Which terminal emulator do you want to use with your Git Bash?

- (●) **Use MinTTY (the default terminal of MSYS2)**

  Git Bash will use MinTTY as terminal emulator, which sports a resizable window, non-rectangular selections and a Unicode font. Windows console programs (such as interactive Python) must be launched via `winpty` to work in MinTTY.

- ( ) **Use Windows' default console window**

  Git will use the default console window of Windows ("cmd.exe"), which works well with Win32 console programs such as interactive Python or node.js, but has a very limited default scroll-back, needs to be configured to use a Unicode font in order to display non-ASCII characters correctly, and prior to Windows 10 its window was not freely resizable and it only allowed rectangular text selections.

8 **Choose the default behavior of `git pull`**

What should `git pull` do by default?

- ( ) **Fast-forward or merge**

  Fast-forward the current branch to the fetched branch when possible, otherwise create a merge commit.

- ( ) **Rebase**

  Rebase the current branch onto the fetched branch. If there are no local commits to rebase, this is equivalent to a fast-forward.

- (●) **Only ever fast-forward**

  Fast-forward to the fetched branch. Fail if that is not possible.
  This is the standard behavior of `git pull`.

9 **Choose a credential helper**

Which credential helper should be configured?

- (●) **Git Credential Manager**

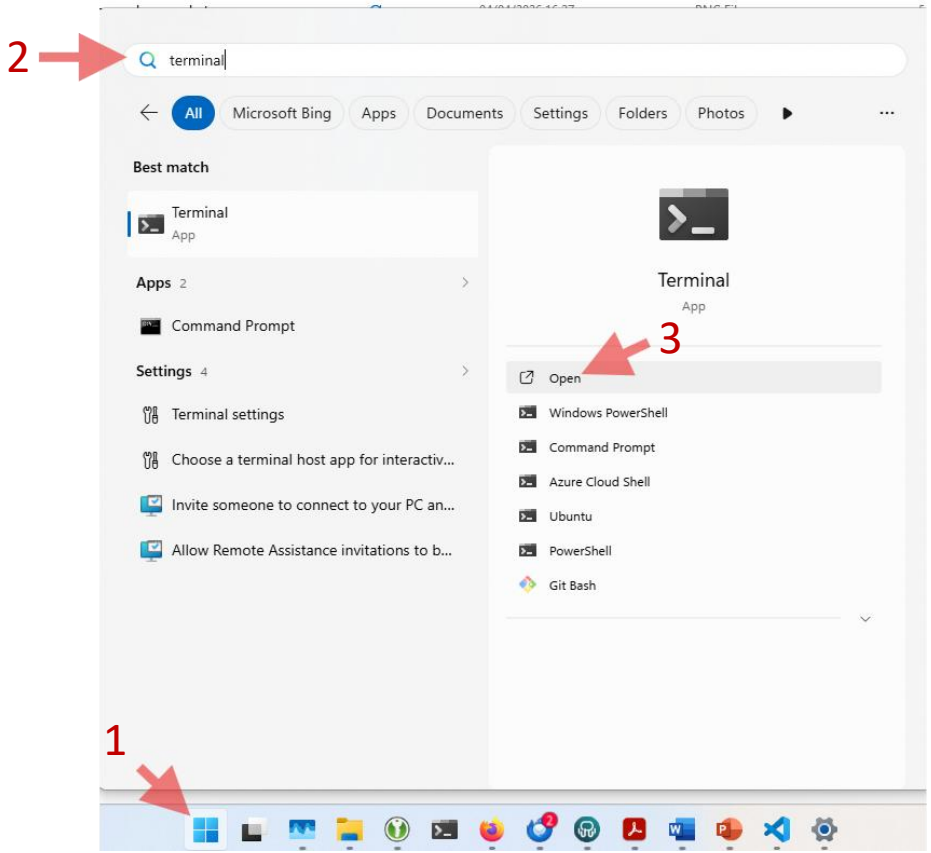
  Use the cross-platform Git Credential Manager.
  See more information about the future of Git Credential Manager here.

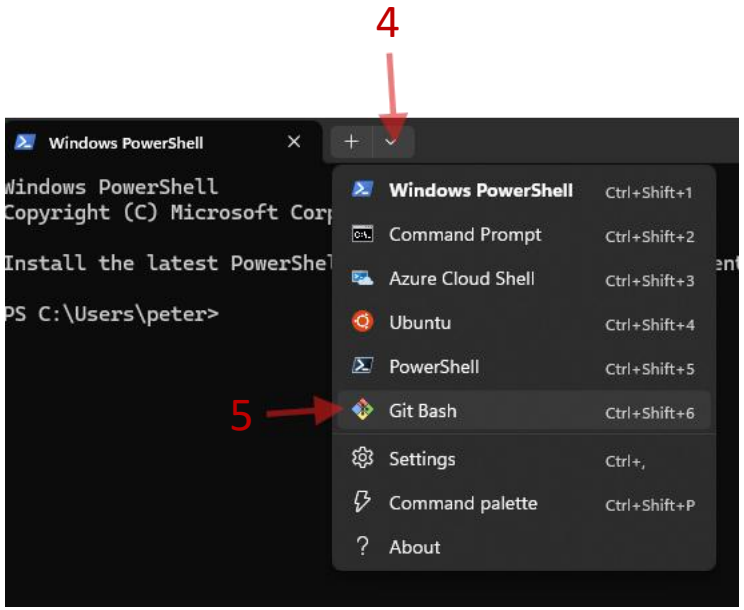
- ( ) **None**

  Do not use a credential helper.

Windows-Terminal

- ✓ Windows-Terminal mit *Git Bash Profile* verwenden
  - deutlich komfortabler als das mitgelieferte Git Bash Terminal





Git Konfiguration

- ✓ In Bash:

```
$ git config --global user.name "<Vorname Nachname>"
$ git config --global user.email "<name@domain.net>"
$ git config --global core.excludesfile 'C:\Users\<username>\.gitignore'
```

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2.4. Python und Poetry

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pyenv-win installieren

- ✓ In Bash:

```
$ git clone https://github.com/pyenv-win/pyenv-win.git $HOME/.pyenv
```

- ✓ *PATH* anpassen
  - *Win + I* und nach *Edit environment variables for your account* suchen
  - User Variable *Path* ändern
  - Neue Einträge *ganz oben* in der Liste einfügen:

```
%USERPROFILE%\.pyenv\pyenv-win\bin
%USERPROFILE%\.pyenv\pyenv-win\shims
```

  - Danach Terminal-Fenster neu starten

Python-Version installieren

✓ In Bash:

```
$ pyenv install 3.14
$ pyenv global 3.14
$ pyenv versions
$ python -version
$ which -a python
```

Poetry installieren

- ✓ In Bash:

```
$ curl -sSL https://install.python-poetry.org | python
```

- ✓ PATH anpassen
  - Neuer Eintrag:

```
%USERPROFILE%\AppData\Roaming\Python\Scripts
```

  - Danach Terminal-Fenster neu starten

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2.5. AeS Repository

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Git Repository klonen und aktivieren

- ✓ In Bash:

```
$ cd /c/my/local/path/to/lecture/material
$ git clone https://github.com/ploh/aes.git aes
$ cd aes
$ poetry sync --only main
$ poetry env info
```

- ✓ In VSCode:
  - File -> Open Folder
  - *Ctrl + Shift + P* -> "Python: Select Interpreter"